

SCHEME OF INSTRUCTION AND EXAMINATION
B. E (AIML)
SEMESTER- IV

SNo	Code	Course Title	Scheme of Instruction			Contact Hrs/Wk	Scheme of Examination			Credits
			L	T	P		Hrs	CIE	SEE	
Theory										
1	MT401BS	Engineering Mathematics – IV (Numerical Methods)	3	0	-		3	40	60	3
2	AI 401PC	Artificial Intelligence	3	0	-		3	40	60	3
3	AI402PC	Computer Organization and Microprocessors	3	0	-		3	40	60	3
4	AI 403PC	Principles of Programming Languages	3	0	-		3	40	60	3
5	AI 404PC	Software Engineering	3	0	-		3	40	60	3
6	ES 402 EC	Signals and Systems	3	0	-		3	40	60	3
7	Professional Elective-I									
	AI 411 PE	Data Visualization	3	0	-		3	40	60	3
	AI 412 PE	Graph Theory								
	AI 413 PE	Microcontrollers and Embedded C								
	AI 414 PE	Operations Research								
	AI 415 PE	Mobile Application Development								
Practicals										
8	AI451PC	Artificial Intelligence Lab	-	-	2		3	25	50	1
9	AI452PC	Computer Organization and Microprocessors Lab	-	-	2		3	25	50	1
10	AI453PC	R Workshop	-	-	2		3	25	50	1
Total			21	0	6		30	355	570	24

MT401BS	ENGINEERING MATHEMATICS –IV (NUMERICAL METHODS)				
Prerequisites		L	T	P	C
		3	0	-	3
Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives :

1	Apply general methodology to solve linear first order and second order partial differential equations
2	To study the classification of second order partial differential equations and solve them by using separation of variables methods
3	To introduce a few numerical methods to solve nonlinear algebraic and transcendental equations and system of linear equations
4	To provide the necessary basic concepts of numerical differentiation, numerical integration
5	To solve the Initial Value Problems.

Course Outcomes :

On completion of this course, the student will be able to :

CO1	Find the solutions of first and second order PDE
CO2	Find solutions of the heat equation, wave equation, and the Laplace equation subject to boundary conditions
CO3	Solve nonlinear equations, system of linear equations
CO4	Find Numerical Integration
CO5	Perform numerical differentiation

UNIT – I

Definition of Partial Differential Equations, First order partial differential equations, Solutions of first order linear PDEs , Solution to homogenous and non-homogenous linear partial differential equations of second order by complimentary function and particular integral method.

UNIT – II

Second-order linear equations and their classification, Initial and boundary conditions, D'Alembert's solution of the wave equation, Heat diffusion and vibration problems, Separation of variables method to Solve simple problems in Cartesian coordinates. The one dimensional diffusion equation and its solution by separation of variables.

UNIT – III

Bisection method, Newton-Raphson method, Solution of linear system of equations- Gauss elimination method, LU decomposition method, Gauss-Jacobi and Gauss-Seidel iteration methods.

UNIT – IV

Interpolation, Lagrange's interpolation, Newton's divided difference interpolation, Newton's Forward and Backward difference interpolations. Numerical differentiation, Interpolation approach, Numerical integration-Trapezoidal rule, Simpson's 1/3 rule.

UNIT – V

Taylor's series method, Euler's method, Picard's method of successive approximations, Runge-Kutta method of 4th order

Suggested Reading:

1	R. K. Jain & S.R.K Iyengar, Advanced Engineering Mathematics, Narosa Publications, 4 th Edition 2014 (Text Book).
2	Erwin Kreyszi, Advanced Engineering Mathematics, John Wiley, 9 th Edition, 2012.
3	B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 43 rd Edition, 2014.
4	M.K.Jain, S.R.K.Iyengar and R.K.Jain, Numerical methods for scientific and engineering computation ,6th Edition , New Age International Limited., 2012
5	B .V . Ramana, Higher Engineering Mathematics, 23 rd reprint, 2015.
6	S.S.Sastry, Introductory Methods of Numerical Analysis, 5th Edition , PHI Private Limited,2012.
7	H.K. Dass, Er. Rajnish Varma, higher Engineering Mathematics, S.Chand Technical 3 rd Edition, 2011.

AI401PC	ARTIFICIAL INTELLIGENCE				
Prerequisites	Introduction to Data Science	L	T	P	C
		3	0	-	3
Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives	
1	To become familiar with basic principles of AI toward problem solving, inference, perception, knowledge representation, and learning.
2	To Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
3	Explore the current scope, potential, limitations, and implications of intelligent systems

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.
CO2	Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.
CO3	Demonstrate an ability to share in discussions of AI, its current scope and limitations, and societal implications.

UNIT – I
Introduction to Artificial Intelligence: Introduction, Brief History, Intelligent Systems, foundations of AI, Sub-Areas of AI, Applications, Tic-Tac Game Playing, Development of AI Languages, Current Trends in AI. Agents: Agents and Environments, Good Behavior: The concept of Rationality, Performance measures, The nature of Environments, The Structure of Agents, Simple agents, Rational agents, problem solving agents, intelligent agents.

UNIT – II
Solving Problem by Searching: Problem-Solving Agents, Searching for Solutions, Uninformed search strategies. Informed Search and Exploration: Informed Search Strategies, Heuristic Functions, Local-Search Algorithms and Optimization Problems. Adversarial Search: Games, Optimal Decisions in Games, Alpha-Beta Pruning, Iterative Deepening.

UNIT – III
Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Knowledge Representation using Frames.

UNIT – IV
Probabilistic Reasoning Expert System: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert System versus Traditional Systems, Rule-Based Expert Systems. Uncertainty Measures: Introduction, Probability Theory, Bayesian Belief Networks.

Fuzzy Logic Systems: Introduction, Crisp Sets, Fuzzy Sets, Fuzzy Terminology, Fuzzy Logic Control, Neuro Fuzzy Systems.
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UNIT – V

Connectionist Models: Introduction: Hopfield Networks, learning in Neural Networks, Applications of Neural Networks, Recurrent Networks, Distributed Representations, Connectionist AI and Symbolic AI. Chatbots

Suggested Reading:

1	“Artificial Intelligence” Saroj Kaushik, Cengage Learning, 2011
2	“Artificial Intelligence” Elaine Rich, Kevin Knight, Shivashankar B Nair, Tata McGraw Hill, Third Edition, 2019
3	“Artificial Intelligence-A Modern Approach” Second Edition, Stuart Russell, Peter Norvig. Create Space Independent Publishing Platform, 2016

AI 403 PC	PRINCIPLES OF PROGRAMMING LANGUAGES				
Prerequisites	Programming for Problem Solving	L	T	P	C
		3	0	-	3
Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives	
1	To introduce the major programming paradigms, and the principles and techniques involved in design and implementation of modern programming languages
2	To introduce notations to describe syntax and semantics of programming languages.
3	To analyze and explain behavior of simple programs in imperative languages using concepts such as binding, scope, control structures, subprograms and parameter passing mechanisms.
4	To introduce the concepts of ADT and object oriented programming for large scale software development
5	To introduce the concepts of concurrency control and exception handling.

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Understand the programming paradigms of modern programming language
CO2	Describe syntax and semantics of programming languages
CO3	Analyze the behavior of simple programs in imperative languages.
CO4	Understand the concepts of ADT and object oriented programming for large scale software development
CO5	Understand the concepts of functional programming and logic programming

UNIT – I
Preliminary Concepts: Reasons for Studying Concepts of Programming Languages, Programming Domains, Language Evaluation Criteria, Influences on Language Design, Language Categories, Language Design Trade-offs, Implementation Methods, Programming Environments, Evolution of the Major Programming Languages. Describing Syntax and Semantics: General Problem of Describing Syntax, Formal Methods of Describing Syntax, Attribute Grammars, Describing the Meaning of Programs.

UNIT – II
Names, Binding, Type Checking, and Scopes: Names, Variables, The Concept of Binding, Type Checking, Strong Typing, Type Compatibility, Scope, Scope and Lifetime, Referencing Environments, Named Constants. Data Types: Primitive Data Types, Character String Types, User- Defined Ordinal Types, Array Types , Associative Arrays, Record Types, Union Types, Pointer and Reference Types, optional types Expressions and Assignment Statements: Arithmetic Expressions, Overloaded Operators, Type Conversions, Relational and Boolean Expressions, Short-Circuit Evaluation, Assignment Statements, Mixed- Mode Assignment

UNIT – III
Statement-Level Control Structures: Selection Statements, Iterative Statements, Unconditional Branching, Guarded Commands. Subprograms: Fundamentals and Design Issues for Subprograms, Local Referencing Environments, Parameter –Passing Methods, Parameters That are Subprograms Names, Overloaded Subprograms, Generic Subprograms, Design Issues for Functions, User-Defined Overloaded Operators. Implementing Subprograms: The General Semantics of Calls and Returns, Implementing “Simple” Subprograms, Implementing Subprograms with Stack-Dynamic Local Variables, Nested Subprograms,

Blocks, Implementing Dynamic Scoping.

Abstract Data Types: The Concept of Abstraction, Introduction to Data Abstraction, Design Issues for Abstract Data Types, Language Examples, Parameterized ADT, Encapsulation Constructs, Naming Encapsulation.

UNIT – IV

Object Oriented Programming: Design Issues, Object Oriented Programming in Smalltalk, C++, Java, C#, Ada 95, Ruby, The Object Model of JavaScript, Implementation of Object Oriented Constructs.

Concurrency: Subprogram level Concurrency, Semaphores, Monitors, Message Passing, Ada Support for Concurrency, Java Threads, C# Threads, Statement-Level Concurrency. Exception Handling and **Event**

Handling: Introduction to Exception Handling, Exception Handling in Ada, C++ and Java, Introduction to Event Handling, Event Handling with Java.

UNIT – V

Functional Programming Languages: Introduction, Mathematical Functions, Fundamentals of FPL, LISP, Introduction to Scheme, COMMON LISP, ML, Haskell, Application of Functional Programming Languages and A Comparison of Functional and Imperative Languages, Functional interfaces (Java 8.0)

Logic Programming Languages: Introduction to Predicate Calculus, Predicate Calculus and Proving Theorems, An Overview of Logic Programming. The Origins, Basic Elements and Deficiencies of Prolog, Applications of Logic Programming.

Scripting Languages: Common Characteristics, Data Types, Object Orientation Names and Scopes, String and Pattern Manipulation Problem Domains, Scripting the World Wide Web

Suggested Reading:

1	“ <i>Concepts of Programming Languages</i> ” Robert.W.Sebesta 12 th Edition, Pearson Education, 2019
2	“ <i>Programming Language Pragmatics</i> “ Michal Scott 4 th Edition Morgan Kaufmann Publishers, 2015
3	<i>Java Precisely</i> Peter Sestoft 3rd Edition, MIT press 2016
4	<i>Programming Languages: Principles & Practices</i> Kenneth A. Lambert and Kenneth C. Loudon, 3 rd Edition, Cengage Learning 2012.
5	<i>Programming languages</i> , Watt, Wiley Dreamtech, First Edition, 2004.

AI 404 PC	SOFTWARE ENGINEERING					
Prerequisites			L	T	P	C
			3	0	-	3
Evaluation	CIE	40 Marks	SEE		60 Marks	

Course Objectives	
1	To introduce the basic concepts of software development- processes from defining a product to shipping and maintaining that product
2	To impart knowledge on various phases , methodologies and practices of software development
3	To understand the importance of testing in software development and study various testing strategies and software quality metrics.

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Acquire working knowledge of alternative approaches and techniques for each phase of software development
CO2	Acquire skills necessary for independently developing a complete software project
CO3	Understand the practical challenges associated with the development of a significant software system.

UNIT – I: Introduction to Software Engineering:

A generic view of Process: Software Engineering, Process Framework, CMM Process Patterns, Process Assessment.

Process Models: Prescriptive Models, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Specialized Process Models, The Unified Models, Personal and Team Process Models, Process Technology, Product and Process.

An Agile view of Process: Introduction to Agility and Agile Process, Agile Process Models.

UNIT – II

Software Engineering Principles: SE Principles, Communication Principles, Planning Principles, Modeling Principles, Construction Principles, Deployment. System Engineering: Computer-based Systems, The System Engineering Hierarchy, Business Process Engineering, Product Engineering, System Modeling.

Requirements Engineering: A Bridge to Design and Construction, Requirements Engineering Tasks, Initiating Requirements Engineering Process, Eliciting Requirements, Developing Use-Cases, Building the Analysis Model, Negotiating Requirements, Validating Requirements.

UNIT – III

Building the Analysis Model: Requirements Analysis Modeling Approaches, Data Modeling Concepts, Object-Oriented Analysis, Scenario-based Modeling, Flow-oriented Modeling, Class based Modeling, Creating a Behavioral Model.

Design Engineering: Design within the context of SE, Design Process and Design Quality, Design Concepts, The Design Model, Pattern-based Software Design..

UNIT – IV

Creating an Architectural Design: Software Architecture, Data Design, Architectural Styles and Patterns, Architectural Design, Assessing Alternative Architectural Designs, Mapping Data Flow into a Software Architecture.

Modeling Component-Level Design: Definition of Component, Designing Class-based Components, Conducting Component-level Design, Object Constraint Language, Designing Conventional Components.

Performing User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation.

UNIT – V

Software Quality Assurance: Basic Elements, Tasks, Goals and Metrics, Formal Approaches, Statistical Software Quality Assurance, Software Reliability, ISO 9000 Quality Standards, SQA Plan.

Testing Strategies: A Strategic Approach to Software Testing, Strategic Issues, Test Strategies for O-O Software, Validation Testing, System Testing, The Art of Debugging.

Testing Tactics: Software Testing Fundamentals, Black-box and White-box Testing, Basis Path Testing, Control Structure Testing, O-O Testing Methods, Testing Methods applicable on the Class Level, Inter Class Test Case Design, Testing for Specialized Environments, Architectures and Applications, Testing Patterns.

Product Metrics: Software Quality, A Framework for Product Metrics, Metrics for the Analysis Model, Metrics for the Design Model, Metrics for Source Code, Metrics for Testing, Metrics for Maintenance.

Suggested Reading:

1	Roger S.Pressman, <i>Software Engineering: A Practitioner's Approach</i> , 7 th Edition, McGraw Hill, 2009.
2	Ali Behforooz and Frederick J.Hudson, <i>Software Engineering Fundamentals</i> , Oxford University Press, 1996.
3	Pankaj Jalote , <i>An Integrated Approach to Software Engineering</i> , 3 rd Edition, Narosa Publishing House, 2008.

AI 412 PE	GRAPH THEORY				
Prerequisites	Data Structures		L	T	P
			3	0	-
Evaluation	CIE	40 Marks	SEE		60 Marks

Course Objectives	
1	To familiarize a variety of different problems in Graph Theory
2	To learn various techniques to prove theorems
3	To understand and analyse various graph algorithms

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Write precise and accurate mathematical definitions of objects in graph theory
CO2	Validate and critically assess a mathematical proof
CO3	Develop algorithms based on diverse applications of Graphs in different domains

UNIT – I
Preliminaries: Graphs, isomorphism, subgraphs, matrix representations, degree, operations on graphs, degree sequences Connected graphs and shortest paths: Walks, trails, paths, connected graphs, distance, cut vertices, cut-edges, blocks, connectivity, weighted graphs, shortest path algorithms Trees: Characterizations, number of trees, minimum spanning trees.

UNIT – II
Special classes of graphs: Bipartite graphs, line graphs, chordal graphs Eulerian graphs: Characterization, Fleury's algorithm, chinese-postman-problem.

UNIT – III
Hamilton graphs: Necessary conditions and sufficient conditions Independent sets, coverings, matchings: Basic equations, matchings in bipartite graphs, perfect matchings, greedy and approximation algorithms.

UNIT – IV
Vertex colorings: Chromatic number and cliques, greedy coloring algorithm, coloring of chordal graphs, Brook's theorem Edge colorings: Gupta-Vizing theorem, Class-1 graphs and class-2 graphs, equitable edge-coloring.

UNIT – V
Planar graphs: Basic concepts, Eulers formula, polyhedrons and planar graphs, characterizations, planarity testing, 5-color-theorem. Directed graphs: Out-degree, in-degree, connectivity, orientation, Eulerian directed graphs, Hamilton directed graphs, tournaments

Suggested Reading:

1	F.Harry, Graph theory, Narosa Publications, 1988.
2	C.Berge: Graphs and Hypergraphs, North Holland/Elsevier, 1973
3	J A Bondy and U.S. R Murthy, Graph Theory with Applications, Elsevier Science Ltd, 1976.
4	Douglas B West, Introduction to Graph Theory, Prentice Hall, 2004.

AI 451PC	ARTIFICIAL INTELLIGENCE LAB				
Prerequisites			L	T	P
			0	0	2
Evaluation	CIE	25 Marks	SEE		50 Marks

Course Objectives	
1	Develop the ability to design and implement solutions for both informed and uninformed search problems in Artificial Intelligence (AI)
2	Acquire proficiency in using Prolog to express and reason about knowledge in first-order logic
3	Utilize the Natural Language Toolkit (NLTK) and advanced techniques to implement Natural Language Processing (NLP) solutions
4	Select and apply relevant Python libraries to synthesize information and construct supervised learning models
5	Develop a comprehensive case study in a multidisciplinary domain, showcasing the integration of AI techniques to solve complex problems.

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Design and develop solutions for informed and uninformed search problems in AI
CO2	Demonstrate reasoning in first order logic using Prolog.
CO3	Utilize advanced package like NLTK for implementing natural language processing
CO4	Demonstrate and enrich knowledge to select and apply python libraries to synthesize information and develop supervised learning models
CO5	Develop a case study in multidisciplinary areas to demonstrate use of AI

List of Programs
- Write a program to implement Uninformed search techniques: - BFS - DFS - Write a program to implement Informed search techniques - Greedy Best first search - A algorithm - Study of Prolog its facts, and rules. - Write simple facts for the statements and querying it. - Write a program for Family-tree. - Write a program to train and validate the following classifiers for given data (scikit-learn): - Decision Tree - Multi-layer Feed Forward neural network - Text processing using NLTK - Remove stop words - implement stemming - POS (Parts of Speech) tagging - In addition to the above programs, students should be encouraged to study implementations of one of the following - Game bot (Tic Tac toe, 7 puzzle) - Expert system (Simple Medical Diagnosis) - Text classification - Chat bot

AI 453PC	R WORKSHOP				
Prerequisites			L	T	P
			0	0	2
Evaluation	CIE	25 Marks	SEE		50 Marks

Course Objectives	
1	Develop proficiency in R programming fundamentals, from installation and comments to variables, data types, and basic output.
2	Gain command over R's flow control constructs, including if-else statements, loops, and functions
3	Learn to work with diverse data structures such as strings, vectors, matrices, lists, arrays, data frames, and factors in R
4	Acquire skills to create various data visualizations in R, including bar plots, histograms, pie charts, box plots, and strip charts
5	Learn how to read/write data, perform data analysis tasks, calculate statistics, and work with univariate, bivariate, and multivariate graphs

Course Outcomes	
On completion of this course, the student will be able to	
CO1	Attain the ability to comfortably navigate and utilize R, covering essential concepts and programming constructs
CO2	Demonstrate the capability to write efficient code using R's control structures and functions for effective problem-solving
CO3	Show competence in representing and manipulating data using various R data structures, preparing for more complex analyses.
CO4	Showcase proficiency in creating informative data visualizations, allowing effective communication of insights from various types of data
CO5	Develop the expertise to manipulate, analyze, and visualize data, enabling effective decision-making in diverse scenarios

1. Installation of R – studio 2. Getting started with R a. R comments b. R variables and constants c. R data types d. Print output e. R numbers 3. R flow control a. R Boolean expression b. R if ... else c. R while loop d. R for loop e. R break and next f. R repeat loop g. R function 4. R Data structures a. R Strings b. R Vectors

- c. R Matrix
- d. R List
- e. R Arrays
- f. R Data Frame
- g. R Factor
- 5. R Data Visualization
 - a. R Bar plot
 - b. R Histogram
 - c. R Pie chart
 - d. R Box plot
 - e. R Strip chart
 - f. R Plot Function
 - g. R Save Plot
 - h. Color in R
- 6. R Data Manipulation
 - a. R Read and Write CSV
 - b. R Read and Write xlsx
 - c. R Dataset
 - d. R min () and max ()
 - e. R mean, median and mode
 - f. R Percentile
- 7. R Program for univariate graphs
 - a. Categorical
 - b. Quantitative
- 8. R Program for bivariate graphs
 - a. Categorical Vs Categorical
 - b. Quantitative Vs Quantitative
 - c. Categorical Vs Quantitative
- 9. R Program for multivariate graphs
 - a. Grouping
- 10. R Program for maps
 - a. Dot density maps
 - b. Choropleth maps
- 11. R program for time dependent graphs
 - a. Time series
 - b. Dumbbell charts
 - c. Slope graphs
 - d. Area charts.
